DATA SCIENCE PROJECT

PREDICTING REAL ESTATE LISTING SUCCESS AND PRICE OPTIMIZATION USING PROPERTY DESCRIPTIONS

NAME: REDDY ABINAYA
STUDENT ID:7235SYXD

1. PROJECT OVERVIEW

1.1 PROJECT TITLE:

Predicting Real Estate Listing Success and Price Optimization Using Property Descriptions

1.2 SUMMARY OF TOPIC AND BACKGROUND:

The United States residential real estate market generates millions of property listings annually, which can be used to predict residential property listing price and listing status outcome by fusing structured property attributes with NLP-derived signals extracted from agent written listing descriptions. Four datasets are combined primary corpus (SAKIB, POLARTECH) provide structured price/bedroom/sqft data across all US states; two state-specific NLP datasets TEXAS, NEW YORK which has real listing descriptions. The modeling is done exclusively to TEXAS AND NEW YORK rows because if we take the full primary corpus all the rows will have descriptions, which means the NLP features will not receive zero-fills, and if it happens the NLP contribution can be measured, so I will be filtering to TEXAS and NEW YORK only in which each row will get a real NLP features.

1.3 RESEARCH QUESTION:

"Which structural and linguistic features most strongly predict sale price and listing success in TEXAS and NEW YORK, and how can SHAP insights generate actionable recommendations?"

1.4 PROJECT OBJECTIVES:

- ❖ To determine the VADER sentiment polarity, Flesch-Kincaid readability scores, luxury keyword count, YAKE key phrases carry a measurable association with listing price and listing success outcome in the market.
- ❖ Evaluate if the Optuna fine-tuning improves the overall accuracy over default baseline, and which model performs the best.
- ❖ Assess whether the stacking ensemble outperforms the base models on both tasks, which can quantify the model's contribution through Ridge meta-learner coefficient analysis.
- ❖ To quantify how NLP features add prediction accuracy beyond the structural attributes alone.
- ❖ Analyze which structural and NLP features drive a strong predicted price and listing success using SHAP TreeExplainer for tree models and permutation importance for the MLP neural network.

1.5 REFERENCES:

Dorogush, A.V., Ershov, V. and Gulin, A. (2018) 'CatBoost: Gradient boosting with categorical features support', Workshop on ML Systems at NeurIPS 2017. Available at: https://arxiv.org/abs/1810.11363 (Accessed: 27 May 2026).

Hutto, C.J. and Gilbert, E. (2014) 'VADER: A parsimonious rule-based model for sentiment analysis of social media text', Proceedings of the 8th AAAI Conference on Weblogs and Social Media, pp. 216-225. Available at: https://ojs.aaai.org/index.php/ICWSM/article/view/14550 (Accessed: 29 May 2026).

Zhao, H. and Wang, K. (2023) 'Predicting real estate price using stacking-based ensemble learning', American Journal of Information Science and Technology, 7(2), pp. 70–75. Available at: https://doi.org/10.11648/j.ajist.20230702.14 (Accessed: 2 June 2026).

2.PROJECT PLAN:

2.1 7-WEEk SCHEDULE:

TASK	ACTIVITIES	DATES
Setup & data Ingestion Literature Review DataCleaning	Google Colab setup, upload all the 4 datasets from kagglehub, raw CSV audit. Reading papers for literature review.	25 May – 1 June
TX+NY Extraction NLP Feature Extraction Assessment 1 Preparation	Extract TX+NY rows only, and stratified sample to 200-300K. Apply features like Vander/Flesh/Luxury/Yake	2 June – 8 June
EDA & Visualization Baseline Models Assessment 2 preparation	Atleast 3-4 visualization after merging,cleaning,extracting the data from the 4 datasets. Baseline models like CatBoost, RandomForest, MLP	9 June – 15 June
Fine-tuned Models OOF +Stacking	Fine tune all the models and record R^2. 5-fold OOF predictions(regression), train Record Ridge meta-learner coefficients and compare individual vs stacked R^2	16 June – 22 June
Ablation Study	Model trained on structural-only features vs structural+NLP features.	23 June – 29 June
SHAP & Interpretability	SHAP TreeExplainer on tuned model, report top-15 features per model.	30 June – 6 July
Report & Assessment 3 submission	Abstract and conclusions, incorporate feedback, Harvard references, GitHub tag. Final submission.	6 July – 10 July

2.2 GANTT CHART

TASK	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7
COLAB SETUP AND KAGGLE DOWNLOAD STATE_MAP NORMALIZATION FIX	●						
LITERATURE REVIEW	●	●	●	●	●	●	●
REPORT WRITING		●	●	●	●	●	●
DATA CLAEANING & IQR/MICE	●	●					
TX+NY EXTRACTION &TARGET ENCODING		●					
NLP FEATURE(VADER/FLESCH/LUXURY/YAKE)		●					
EDA			●				
ASSESSMENT 1 SUBMISSION		♦	●				
BASELINE MODELS(RF/CatBoost/MLP)				●			
OPTUNA FINE-TUNNING (ALL 3 MODELS)				●			
5-FOLD OOF+ RIDGE STACKING					●		
ABLATION STUDY					●	●	
SHAP +PERMUTATION IMPORTANCE						●	
ASSESSMENT 2 SUBMISSION				♦		●	
REPORT FINALIZATION AND GITHUB TAG						●	
FINAL SUBMISSION							♦

3. Data Management Plan

• Overview of the dataset:

Dataset Name	Size	States	Key Columns Used	Has Desc?	Role
SAKIB (ahmedshahriarsakib)	2.2M rows × 10 cols	All 50 (TX+NY extracted)	price, bed, bath, house_size, acre_lot, state, status	No	TX+NY struct backbone
POLARTECH (polartech)	~500K rows × 28 cols	All 50 (TX+NY extracted)	price, bedrooms, bathrooms, sqft, state, city	No	TX+NY struct supplement
TEXAS 2026 (jahnavikachhia23)	~12K rows × 13 cols	TX only	description, price, sqft, bedrooms, bathrooms, home_type	**Yes**	NLP profile source — TX
NEW YORK 2026 (kanchana1990)	~9K rows × 11 cols	NY only	description, price, sqft, beds, baths, status	**Yes**	NLP profile source — NY

• Data Collection:

All the datasets are downloaded using the kagglehub link because the size of the datasets are large if we do manual CSV uploads it will take more time.

• Metadata:

Attribute	Details
File formats	Raw data: CSV (.csv). Processed through kagglehub link. Code: .py and .ipynb. Report:.pdf.
Estimated sizes	Raw: SAKIB ~300MB, POLARTECH ~190MB. TEXAS ~11.5MB, NEW YORK ~9MB Processed CSV: ~100-300MB. Code and notebooks: ~3-5MB. Total project storage: ~600MB-1GB.
No PII	None of the four datasets contains names, contact details, financial account numbers, or any data capable of identifying a property owner.

• Version Control:

- GitHub repository: https://github.com/7235SYXD/Real-Estate.git.
- Minimum one commit per week with clear messages.
- File naming in snake_case and version tags (v1.0, v2.0 etc.) will be used.

• ReadMe File:

- A detailed README.md will be added in Week 7.
- It will include project overview, research question, setup instructions, run steps, limitations, and references.

• Security & Storage

- Primary: Public GitHub.
- Secondary storage: OneDrive.

• Ethics:

- No PII present in any dataset. All data publicly available on Kaggle for academic use.
- Geographic bias will be analyzed and reported. Full transparency via public GitHub.